

enced significant enhancements. For all production processes, LED manufacturers were using class 10,000 clean room. Now, they have shifted to class 1000 clean room, which means that particles larger than half micrometre are not allowed in the space where LEDs are manufactured. Earlier, in class 10,000 clean room, up to 70 such particles could enter the space. Consequently, the structural efficiency of LEDs has increased to about 90 per cent, which, in turn, provides quality visual performance.

Enhanced processes. To provide better visual performance, red and blue dyes are fused in the same package coated with phosphor. This improvement in the manufacturing process creates a visually appealing warm white light.

High-grade materials. Today, the wafers used to create LED dyes have doubled in size. Over a decade ago, seven-to-ten-centimetre wafers were used to manufacture LEDs, whereas today 15cm wafers are used that produce up to two times the volume of LEDs, resulting in cost-saving of about 30 to 50 per cent.

Innovative applications

The initial idea for smart lighting was remotely-controlling lighting, which includes dimming and switching it on/off. Today, there are a lot of other features being developed and incorporated. Bhaskar informs, "One such feature is occupancy-sensor-controlled lighting, where the entire room or certain section of it is lighted-up when a person enters and automatically turns off as the person exits the room." Thawani adds, "Another feature, time-scheduled light control, helps gradually dimming the light as the day progresses. This can be achieved with the help of a microcontroller."

Different lights in a room give different light-intensity outputs, according to the preferred mood-lighting presets. Jitin says, "Lights in a room change colour and intensity according to the beat of the music in that room, and the most interesting thing is that, all these are happening accurately with

Advantages of adopting smart lighting

Smart lighting systems deliver operational control, improved light and, of course, big financial savings. Factors driving the adoption of smart lighting systems at home, office or industrial units are:

Massive energy reduction. High energy savings can be achieved by carefully engineering the delivery of light by providing it only when required. By incorporating advanced LED technology, smart lighting systems can pull down lighting costs by a massive 90 per cent, when compared to traditional alternatives. Thawani says, "LED is the most eco-friendly technology in lighting till date."

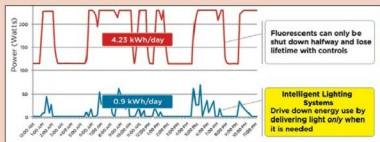


Image courtesy: <http://www.digitalumens.com>

More light, where required. The directional and efficient light supplied by LED based fixtures can significantly boost light levels and uniformity of lighting in a residential, commercial or industrial space.

Long life, reduced maintenance. With a lifespan of up to 100,000 burning hours, LED light sources used in high-quality smart lighting systems virtually get rid of regular re-lamping schedules and ongoing maintenance. When coupled with an occupancy sensor and intelligent control, reduced run hours additionally increase useful life.

Improved operational control. The ability to immediately update the lighting program, merged with the ability to collect detailed data about a building's or area's energy consumption, lighting delivery, occupancy and other parameters mean that smart lighting systems give managers better grip over their resources and working environment.

Sustainability. Corporations, especially, are enthusiastically accepting smart lighting systems to decrease the load on their utilities, document efficiency improvements, reduce carbon footprint and safeguard themselves from future rate increases.

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customisable preset recording or with a soft touch of your screen from your smartphone or tablet."

Additionally, smart lighting can adjust from dark to light in a more gradual manner than conventional lighting. Jaiwin says, "This gives older eyes time to adjust and reduces undue strain. This saves energy and is safer to boot."

All in all, smart lighting is very promising

Investing in intelligence rewards dividends. If you are considering a new installation or retrofit, developing a thor-

ough understanding of smart lighting systems and opportunities they hold in store for greater operational and energy-efficiency is worthwhile. Lighting is usually the largest energy user in industrial, residential or commercial space, but is absolutely the easiest to address with smart lighting. It is time to get out your utility expenditure, start to assess your current lighting costs incurred every month and how far you can reduce that number with a smart lighting system. ●

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